

1 HYDRA-BDE Input Types (relating to operations)

1.1 Serial production

The serial production is the "classic" approach of manufacturing operations. This means that only one operation is produced at a machine at a time.

Serial production has the advantage that machine efficiency can be easily planned.

1.2 Parallel production

Parallel production means that more than one operation is processed simultaneously on the same machine.

At **individual workplaces** the system assumes that the two operations belong together (e.g. production of upper and lower parts with the same tool as two separated operations). Consequently, the system connects each person (who is logged on) with every operation. However, the labor utilization relating to operations is thereby halved.

At **group workplaces**, in contrast, parallel production is possible without the system linking the person to all operations that are currently being processed. This is mainly used at assembly workstations where several workers at the "Assembly" workstation process several operations in parallel. A disadvantage of this option is that the planning of such workstations is difficult.

1.3 Merged operations

This is a special type of serial production. At the planning stage (e.g. HYDRA shop floor scheduling) or at the shop floor terminal, different short running operations are grouped together in logical blocks with an easy to handle running time (i.e. merged operations). For these merged operations HYDRA creates a "representative" operation, which is logged on for all individual operations included. The entered data is divided according to different configurable perspectives.

1.4 Splits

If an operation should be processed on several machines in parallel, the HYDRA shop floor scheduling module allows for the operation to be split into several "splits". These splits are handled by HYDRA like separate operations and can separately be logged on and off independently of each other. The collection of all actual data then pertains to the particular splits and to the original operation (master split), which is known to the PPS system.

1.5 Multiple machine production

If no planning tool is used in HYDRA (e.g. HYDRA shop floor scheduling module), the system can be configured in a way that allows for an operation to be processed, i.e. logged on, on several machines at the same time, without having to be split. Data is collected separately for every machine. A disadvantage of this variant is that planning becomes very difficult.

1.6 "Mixed operation"

As regards the shop floor, HYDRA generally supports all forms of mixed operation. Consequently, splits, merged operations and "normal" operations can be processed serially, in parallel or as multi-machine production. Data collection refers in each case to the collection forms presented above.

In this way, it might be possible in HYDRA, for example, that while an upper and lower part are produced, the operation called "compression lower part" can be logged on to another workstation at the same time for reworking purposes (e.g. debur).